

Stagnone di Marsala digital twin: the seagrass parametrisation was not reaching the model

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To: Prof. Giuseppe Ciraolo, Prof. Mauro De Marchis

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1. Two faults in the seagrass parametrisation

My report of 8 August concluded that distributed seagrass roughness is a third-order refinement, changing Lagrangian skill by 0.009 or less in three of its four conditions. That conclusion was wrong, and the reason is now established. Two separate faults meant the seagrass was never actually resisting the flow, and once both are repaired the roughness term becomes the strongest single effect in the ensemble rather than the weakest.

The first fault was in the input file. D-Flow FM matches the roughness area file to the mesh with a tolerance of 1 cm, and our coordinates were written with six decimals. That is about 11 cm at this latitude, so 94% of the seagrass links were silently discarded. The model was reporting a distributed roughness field of which it had applied 5.5%.

The second fault is in the solver, and it is not ours. The Baptist formulation splits vegetation resistance into a bed roughness and a canopy momentum sink, and the D-Flow FM technical reference states that the sink enters the momentum equation. It does not. The routine computes it and the solver never reads it. I established this by running two identical simulations whose only difference was the canopy drag coefficient, zero against 0.8, giving a momentum sink of 0 m^{-1} against 4 m^{-1} over 16359 links. The two solutions are bit-identical across 48 output steps and 27901 cells. The formulation had been delivering only half of itself, and the half it delivered made the meadow smoother than bare sand.

Both faults look identical from the outside, which is why they survived so long. The configuration file is accepted, the diagnostic echoes every parameter back correctly, and the simulation runs to completion.

2. The parametrisation now used, and how it was fixed

Flexible Mesh carries a second, independent vegetation implementation that is separate from the roughness machinery, and there the canopy term does reach the momentum equation. I rebuilt the four vegetated members on it. Every parameter comes from published measurement, and none was fitted to any observation:

Parameter	Value	Source
Shoot density	500×6 leaves, times cover	Ciraolo et al. (2006), times the satellite classification
Leaf width	0.01 m	Ciraolo et al. (2006)
Canopy height	0.40 m	Prinos et al. (2010), submergence 0.32 to 0.50
Leaf density	900 kg m^{-3}	Prinos et al. (2010), natural 800 to 1020
Drag coefficient	0.8	rigid cylinder, not adjusted

The canopy height deserves a note, because I had it wrong first. I took the leaf length from your 2006 study, over 1 m in nature, and used it directly. What the model needs is the vertical extent the drag occupies, and for a plant whose leaves lie over those differ, which is why your paper reports the deflected thickness k separately. Your measured k/h sits mostly between 0.2 and 0.6, and field observation agrees that *Posidonia* here rarely reaches the surface. At 0.40 m the modelled resistance lands within 3% of

your Equation 22, evaluated at the model's own velocities and depths, with nothing tuned against it.

Two checks that were not used to choose any parameter. Surface over depth-averaged speed on the meadow is 1.60, where the roughest bed roughness we can impose reaches 1.335, so the vertical signature is a canopy and not a rough bed. And bed shear stress falls by a factor of 11, whereas an equivalent bed roughness raises it: the meadow shelters the bed, which the previous parametrisation had backwards.

3. What the corrected ensemble shows

The 35 drifters were held out of every one of these decisions.

	Liu–Weisberg skill, and change against the cell’s uniform control			
Configuration	Uniform	Corrected 153	Formula 154	Native module
No waves, fixed bed	0.539	+0.163	−0.065	+0.195
Waves, fixed bed	0.505	+0.208	−0.082	+0.188
No waves, mobile bed	0.469	+0.190	−0.057	+0.256
Waves, mobile bed	0.624	+0.081	−0.016	+0.087

Distributed roughness now helps in all four cells, not one, and the cell where the August report said it acts is the cell where it acts least. Two independent routes agree, the repaired roughness file and the native module, and they differ from each other by 0.03 or less. Formula 154 is negative everywhere, which is the signature of the inert sink rather than a third opinion.

4. This bears directly on the bathymetry question

In August I asked whether to wait for a new survey, on the reasoning that a mobile bed might be helping because it moves an outdated morphology towards the present one, making the improvement a compensation rather than a result. The corrected ensemble gives evidence on exactly that mechanism, and it is not the answer I expected.

With uniform roughness, waves alone reduce skill by 0.035 and a mobile bed alone reduces it by 0.071, the latter making every one of the 35 drifters worse. Yet together they raise it by 0.084. An additive model predicts –0.106, so the interaction term is +0.190. With the seagrass correctly represented, that same interaction term falls to +0.028. The synergy exists only when the meadow is missing.

So the compensation is real, and part of what it compensated for was the absent canopy rather than the old bathymetry. This does not remove the bathymetric concern, but the earlier evidence for the interaction was resting partly on a defect we have since fixed. Standard deviation of skill across the four cells is 0.057 for the uniform members and 0.016 for the vegetated ones: once the meadow is right, transport skill is largely indifferent to whether waves and bed mobility are on.

5. Consequences and what I propose

Sections 4.3 and 5 have to be rewritten, and the central claim changes from an interaction between waves and bed mobility to something I think is stronger: a physically grounded canopy is what the transport prediction needs, and omitting it manufactures a synergy between two other processes.

I would value your view on two points. Whether to present the inert momentum sink as a finding in the paper, with the bit-identical test as evidence, or to report it to Deltares separately and keep the paper on the physics. And whether the title should now foreground the seagrass.

One limitation to record. Your Equation 22 makes the friction factor depend on velocity, and Flexible Mesh offers that only in its depth-averaged solver, while ours is three-dimensional. It does carry a



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mechanical bending model, which I verified is genuinely dynamic, but its amplitude over our velocity range is about 4%, against the factor of twelve your measurements span between 2 and 30 cm s^{-1} .

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